Exercise Sheet 11

Integration — Network Science Across All Topics

5942UE Network Science

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11.A From Structure to Process: A Case Study

Exercise 11.A.1: Network Characterization

Karate club graph — full structural profile.

1. Is the graph connected? What is the diameter?
2. Compute global and average clustering.
3. Top-3 nodes by degree, betweenness, closeness centrality. Same nodes in all three?
4. Are there any bridges?

Exercise 11.A.2: Community and Social Structure

1. Detect communities via greedy modularity. How many? Modularity score?
2. Compare to the known `"club"` faction split. Does it match?
3. Compute the E-I index for the faction.
4. Treat within-community ties as positive, cross as negative — is the network structurally balanced?

Exercise 11.A.3: Ties, Paths, and Information Flow

1. Classify each edge as a strong tie (≥ 2 common neighbours) or weak tie. What fraction is weak?
2. Are any weak ties also bridges? What does this confirm about Granovetter?
3. Compare L and C to an Erdős–Rényi graph with the same n and m. Is karate club a small-world network?

11.B Advanced Dynamics and Robustness

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Exercise 11.B.1: Vaccination Strategy

You can vaccinate (immunize) only 5 of 34 nodes. Compare three strategies:

1. **Random:** 5 random nodes
2. **Hub:** top-5 by degree
3. **Bridge:** top-5 by betweenness

For each: run SIR ($\beta = 0.4$, seed = 2nd-highest-degree node, 50 runs). Average fraction infected?

When does the bridge strategy beat the hub strategy?

Exercise 11.B.2: Cascade vs. Epidemic

Compare on the karate club:

- **Simple contagion:** SIR, $\beta = 0.4$
- **Complex contagion:** threshold cascade, $\theta = 0.35$

For each, seed from node 0 (hub).

1. Which spreads faster to 50% adoption?
2. Which reaches more nodes total?
3. Now seed from node 11 (peripheral, degree 1). Does the comparison change?

Exercise 11.B.3: Full Network Pipeline (Capstone)

Build a complete pipeline that, for any input graph, reports basic stats, clustering, top centrality, communities, modularity, and a quick SIR epidemic from the hub. Run it on:

1. Karate club
2. `nx.barabasi_albert_graph(50, 2)`
3. `nx.watts_strogatz_graph(50, 4, 0.1)`

(a) Which graph is most clustered? Smallest diameter? (b) Which has highest epidemic vulnerability from the hub? (c) Which best models real social networks?

Exercise 11.B.4: Concept Map — Synthesis

No code. Connect concepts across the course:

1. **L1–L2:** Why does connectivity matter for epidemics?
2. **L3–L8:** Granovetter's weak ties are bridges. How do they affect simple vs. complex contagion differently?
3. **L4–L8:** When do high-betweenness nodes matter for vaccination, and when do they not?
4. **L5–L8:** How does homophily affect complex contagion?
5. **L7–L8:** Which small-world property helps simple contagion? Which helps complex contagion?

Wrap-Up

- structure and dynamics interact: same graph, different process rules → different outcomes
- centrality is not a single concept; the right measure depends on the question
- targeted interventions (vaccination, seeding) require matching the strategy to the contagion type
- "small world + clustered communities" is the realistic baseline for most human networks